

Investigating apparent null bias

HPS-GPR study v5.9.5

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Follow-up to slides 14 and 50 of `unblind_meeting_RCmeet`

Result

The positive location of the slide-50 scan-maximum distribution is expected even for a perfectly centered null. There is also a real, mass-dependent offset in the *signed, fixed-mass* response to the frozen generating source. These effects must be separated. The offset has both signs across the mass scan; it is not a uniform upward displacement.

At the observed 2021 peak, 78 MeV, the saved 256 complete Poisson refit scans have signed-root mean 0.548 (point-wise 95% interval $[0.423, 0.672]$), consistent with the deterministic source response $a = 0.577$. The conventional standard-normal tail of the observed root $r_{\text{obs}} = 2.809$ is 0.00249; the marginal tail under the source-response model is 0.01138. This is a local calibration difference before taking a scan maximum. The selected peak is an illustration, not a preselected local test.

The slide-14 statistic is a different object: a squared, covariance-weighted residual summed over window bins. Its expectation need not equal the bin count. The new calculation predicts $\mathbb{E}[Q/N_{\text{bin}}]$ between 0.998 and 1.185 over 50–250 MeV under the frozen source. At 78 MeV the prediction is 1.026, of which only 0.030 is the deterministic bias contribution. Thus the large observed residual feature near 78 MeV is not explained by a large uniform null offset.

What was tested

The primary sample is the released 2021 native 10% dataset, with the production $\pm 2.25\sigma_m$ exclusion window fixed. Slide 14 uses 201 masses at 1 MeV spacing; slide 50 uses 401 masses at 0.5 MeV spacing. Both use the same observed counts and frozen GP generating source. The nominal support is 36–300 MeV; the saved whole-bin edges are 36–299.75 MeV.

The study reuses 256 complete Poisson spectra/refit scans and 200,000 saved Gaussian-field maxima. It adds exact sideband-prediction replays and 200,000 paired Gaussian-field controls. No new signal optimization, production window choice, signal-shape selection, or observed-root recentering is performed.

Quantity	What is expected under a centered null?	What would indicate an offset?
Signed root $r(m)$	Mean zero, if locally calibrated	Nonzero fixed-mass ensemble mean
Positive root $\max(r, 0)$	Positive mean, with an atom at zero	Compare against its clipped law
Scan maximum T	Positive distribution shifted by the search	Compare against the same full-scan null
Residual Q/N_{bin}	Positive; mean one only under extra conditions	Separate variance and mean-residual terms

Claim boundary. All numerical probabilities here are conditional on a frozen source estimated from observed data. Monte Carlo intervals do not include uncertainty in that source. This study does not establish unconditional discovery calibration or confidence-limit coverage.

1 Why a null scan maximum is positive

Write the signed profile-likelihood root as $r(m)$ and the displayed upward excess as $Z(m) = \max(r(m), 0)$. The one-sided likelihood convention produces a boundary atom even when the unconstrained signed root is standard normal [1]. For a centered unit normal R ,

$$\mathbb{E}[R] = 0, \quad \mathbb{E}[\max(R, 0)] = \frac{1}{\sqrt{2\pi}} = 0.399, \quad P(\max(R, 0) = 0) = \frac{1}{2}. \quad (1)$$

For N independent centered unit normals and $T = \max(0, R_1, \dots, R_N)$,

$$P(T \leq t) = \Phi(t)^N \quad (t \geq 0), \quad P(T = 0) = 2^{-N}. \quad (2)$$

The maximum moves upward without any fixed-mass mean bias. The HPS masses are correlated, so $N = 401$ cannot be treated as 401 independent trials. Modeling the correlated significance field is the purpose of the GP trials approach [2].

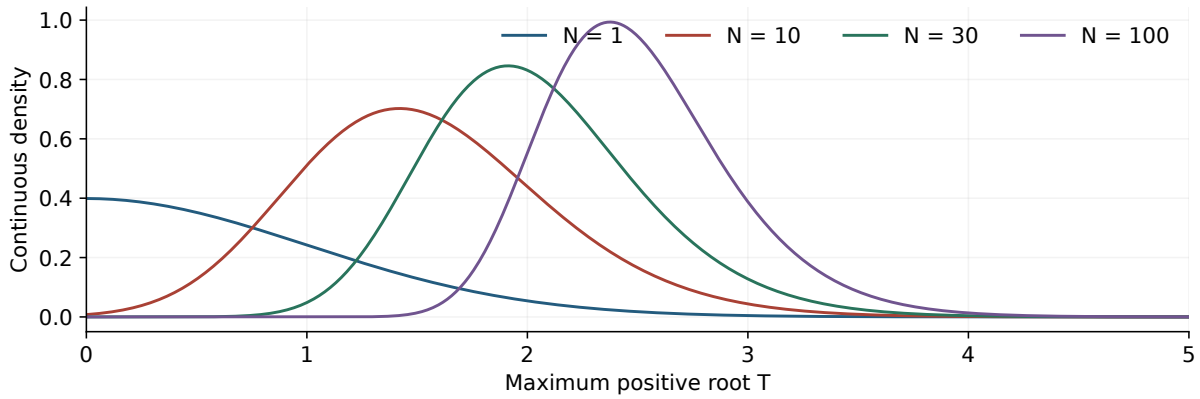


Figure 1: Independent-normal illustration. The shown N values are not fitted HPS effective-trials counts. The continuous density omits the discrete atom at zero.

The actual response approximation retains the mass-dependent mean, scale and correlation:

$$r^*(m) = a(m) + s(m)W(m), \quad W \sim \mathcal{N}(0, R), \quad T^* = \max_m \max(r^*(m), 0). \quad (3)$$

Here $a(m) = r(B; m)$ is the deterministic response to source B , not an assumed identity with the exact Poisson ensemble mean. The stored array named K is R , a unit-diagonal correlation matrix; raw covariance is $\text{diag}(s)R \text{diag}(s)$. This distinction was independently checked.

Three distributions and the local-to-scan transition

A positive maximum is not, by itself, a biased signed estimator

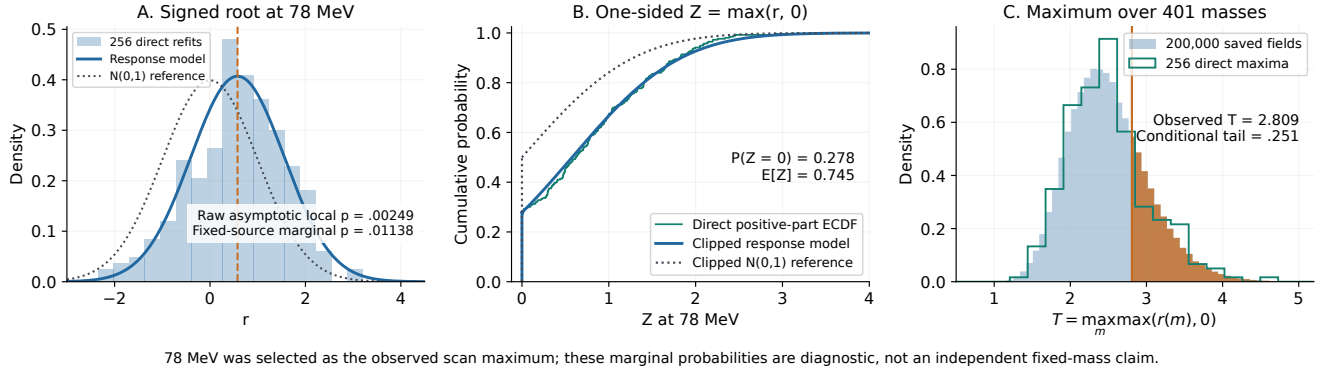


Figure 2: Three distinct distributions under the same source. The observed threshold stays raw. The maximum distribution has median 2.446, although the signed response is close to zero over much of the mass range. The 78 MeV illustration was selected by the observed maximum.

Local calibration and the look-elsewhere step are different

At the selected 78 MeV point,

$$p_{\text{local,asym}} = \bar{\Phi}(r_{\text{obs}}) = 0.0024875, \quad (4)$$

$$p_{B,\text{marginal}} = \bar{\Phi}\left(\frac{r_{\text{obs}} - a}{s}\right) = 0.0113826, \quad (5)$$

$$p_{B,\text{scan}} = P_B(T^* \geq T_{\text{obs}}) = 0.250919. \quad (6)$$

The ratio $p_{B,\text{scan}}/p_{B,\text{marginal}} = 22.04$ compares two tails under the same source model and threshold. Dividing by the conventional asymptotic local tail instead gives 100.87 and mixes local calibration with the search effect. Neither ratio is a universal constant trials factor.

Saved direct refits give 2/256 local exceedances (95% binomial interval $[0.00095, 0.02793]$) and 57/256 scan exceedances ($[0.1732, 0.2786]$). The latter contains the Gaussian-field scan probability. The local count is too small to resolve a precise rare-tail shape. Subtracting a and dividing by s in the observed scan would define a different ordering statistic; this study does not adopt it.

Where the fixed-mass offset comes from

The masked predictor need not reconstruct the generating curve exactly: $\delta = B_w - b(B_t)$ can retain smooth, signed interpolation residuals. With signal template w and frozen residual covariance V_0 , the Gaussian generalized-least-squares approximation projects that residual as

$$a_{\text{GLS}} = \frac{w^\top V_0^{-1} \delta}{\sqrt{w^\top V_0^{-1} w}}. \quad (7)$$

At 78 MeV this gives 0.576797, matching the saved profiled response 0.576784. Thus the offset is already present when the generating source, without Poisson noise, is passed through the masked prediction and signal fit. It is not created by taking a scan maximum. This establishes the mechanism under this source; it does not establish that the source is the physical background.

2 A real fixed-mass response offset, with both signs

Across the 2021 scan, $a(m)$ ranges from -1.096 at 71 MeV to $+1.400$ at 50 MeV. Its mass-grid mean is -0.007 and RMS is 0.251. Averaging over mass therefore hides local offsets. The response scale lies between 0.945 and 0.987. The mean of the direct Poisson refits follows this structure.

2021 native 10% | 50–250 MeV | fixed source and $\pm 2.25\sigma$ window

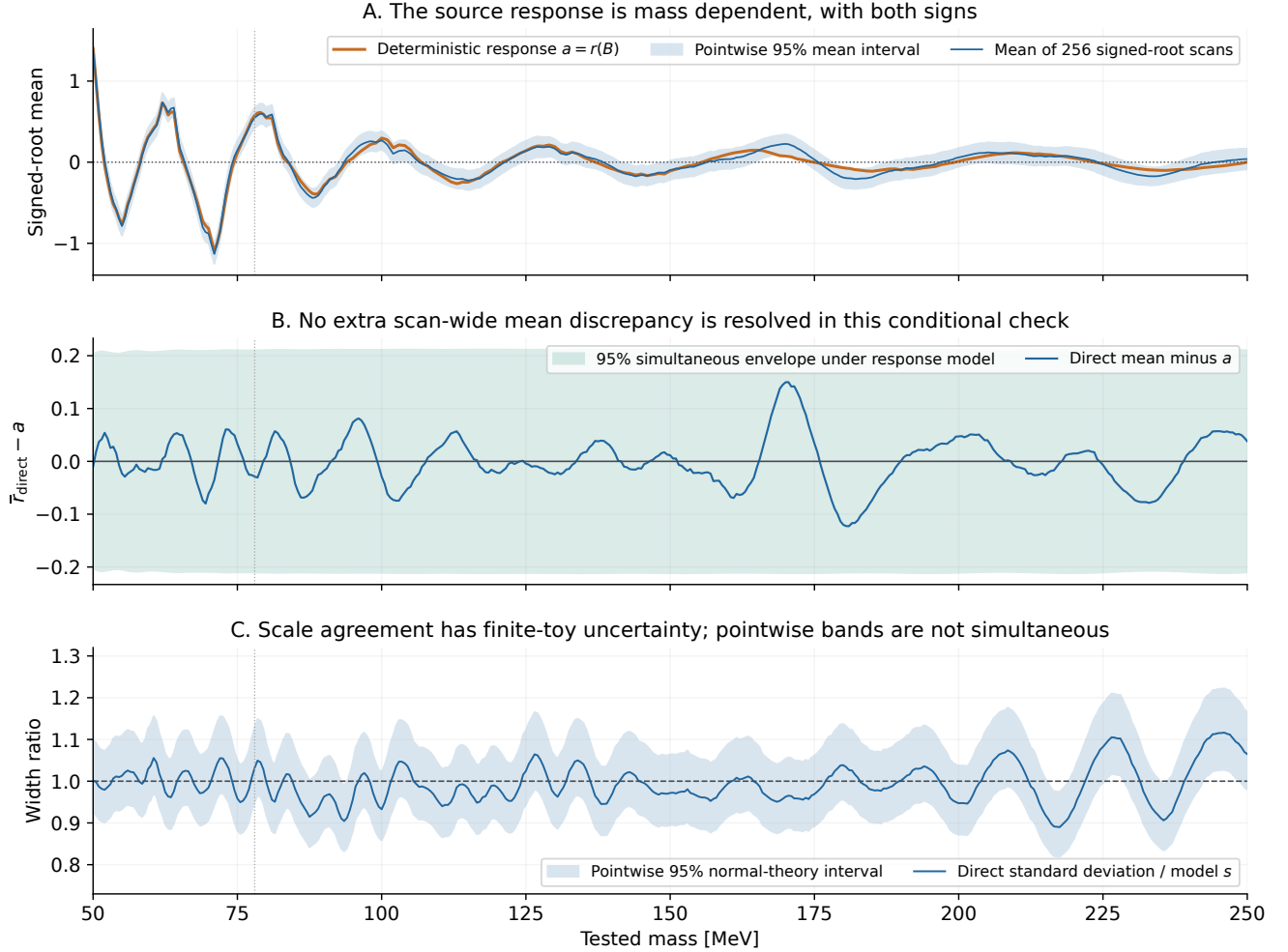


Figure 3: The signed-root mean is the relevant bias diagnostic. Panel B tests only the discrepancy between the complete-refit mean and the response approximation a ; it does not test whether $a = 0$. Pointwise intervals in A and C are not simultaneous. The simultaneous envelope in B assumes the stated Gaussian response model.

For 2021, the largest standardized difference between the direct mean and a is 2.433. The model-based 95% simultaneous threshold is 3.435 and the conditional tail is 0.557. This resolves no additional scan-wide mean mismatch at the available precision. It does not remove the nonzero fixed-mass offset or prove every approximation accurate in rare tails.

3 How much the mean, width and correlations matter

Every control below keeps the observed raw maximum fixed at 2.808645. Paired normal draws change one part of the assumed null at a time. They diagnose the mechanism; they are not alternative production probabilities.

Null control	Median T^*	Tail at T_{obs}	95% MC interval
Nominal a, s, R	2.445	0.25113	[0.2492, 0.2530]
Remove a only	2.291	0.16225	[0.1606, 0.1639]
Set $s = 1$ only	2.492	0.28256	[0.2806, 0.2845]
Set $a = 0, s = 1$	2.338	0.18779	[0.1861, 0.1895]
Independent mass nodes	2.966	0.67394	[0.6719, 0.6760]
Perfectly correlated nodes	1.400	0.06772	[0.0666, 0.0688]

Table 1: Paired Gaussian controls, 200,000 draws per row. Intervals quantify Monte Carlo error within each specified null.

Removing a alone reduces the conditional tail by 0.0889 ± 0.0016 (paired 95% Monte Carlo interval). Even then, the maximum has median 2.291. The upward position of the histogram is mainly a search-maximum effect; the local offsets make a material additional contribution to its tail.

Paired null controls: the observed raw maximum stays fixed at 2.809

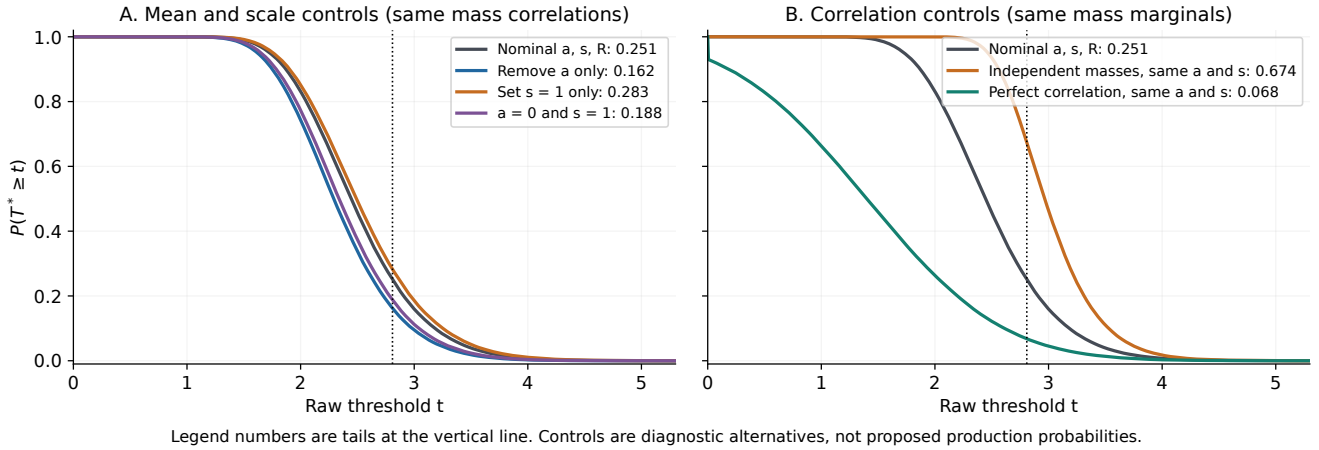


Figure 4: Mean/scale controls retain the same mass correlation. Correlation controls retain the same marginal means and scales. Perfect correlation and independence are diagnostic comparison models, not proposed physical models.

4 What slide 14 measures

For window bins w and disjoint sideband bins t , let $b(Y_t)$ be the GP background prediction and $C_{\text{GP}}(Y_t)$ its posterior count covariance. Slide 14 uses

$$e = Y_w - b(Y_t), \quad V(Y_t) = \text{diag}(b(Y_t)) + C_{\text{GP}}(Y_t), \quad Q = e^\top V^{-1} e. \quad (8)$$

The posterior covariance describes uncertainty under the fitted GP model [3]. It is not automatically the repeated-Poisson covariance of the sideband estimator under a fixed generating spectrum B .

Freeze $V_0 = V(B_t)$, define $\delta = B_w - b(B_t)$ and linearize the estimator with $J = \partial b / \partial Y_t|_{B_t}$. Independence of window and sideband Poisson counts gives

$$W \simeq \text{diag}(B_w) + J \text{diag}(B_t) J^\top, \quad (9)$$

$$\mathbb{E}[Q_0] \simeq \underbrace{\text{tr}(V_0^{-1} W)}_{\text{repeated-sampling variance}} + \underbrace{\delta^\top V_0^{-1} \delta}_{\text{prediction-bias contribution}}. \quad (10)$$

Here $Q_0 = e^\top V_0^{-1} e$ freezes the weighting covariance. For any residual distribution the trace-plus-mean identity is exact with fixed V_0 and the exact residual mean and covariance. The displayed use of δ and W is a first-order approximation. A denominator of N_{bin} is a scale convention, not a fitted number of degrees of freedom.

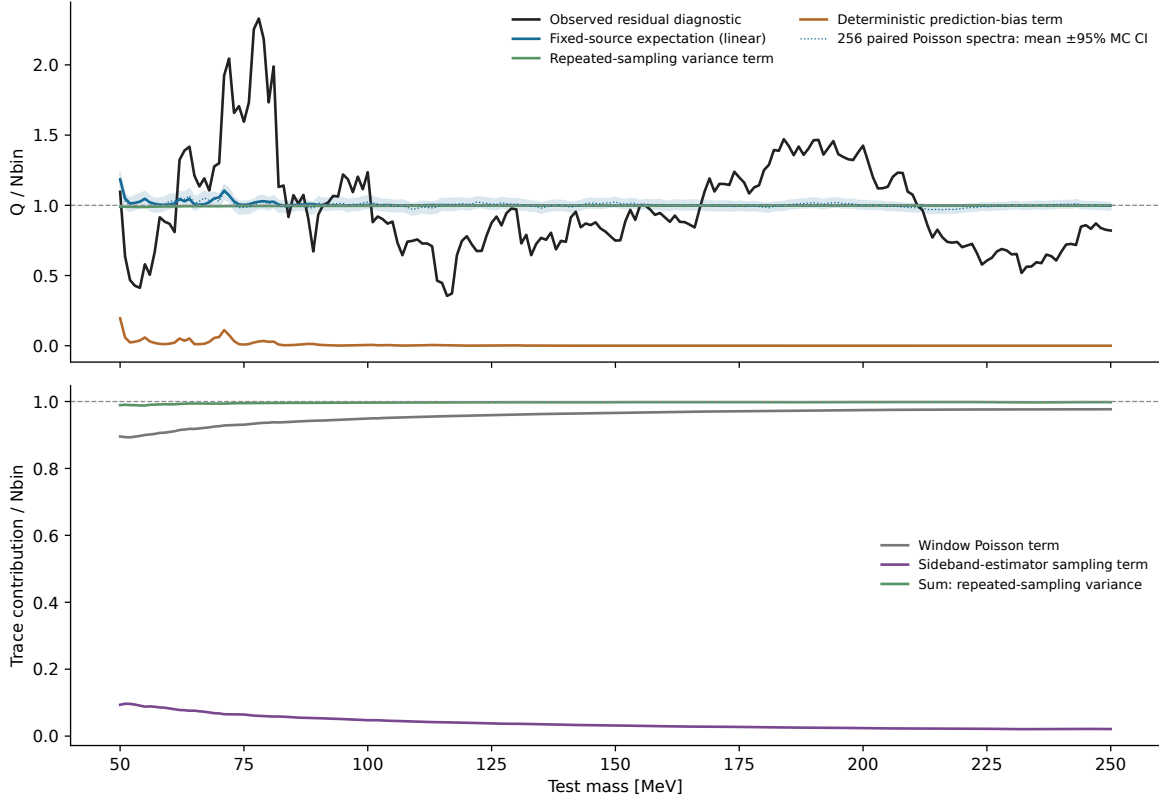


Figure 5: The observed residual curve and the conditional mean answer different questions. The lower panel separates Poisson noise in the excluded window from variation of the sideband estimator. Exact paired replays recompute the GP prediction and covariance on each saved toy.

5 Paired residual controls and the observed feature

The replay differentiates both $\alpha_i = 1/Y_i$ in the log-count GP noise and the lognormal mean correction $\exp(\mu + \frac{1}{2}C_{ii})$. Directional finite differences check the Jacobian; the maximum relative discrepancy is below 10^{-6} . The analytic calculation covers all 201 masses.

For each saved complete Poisson spectrum, paired controls freeze the background prediction, refit the sidebands, remove only the deterministic residual δ , or update the weighting covariance. Exact replay uses the archived kernel parameters and does not optimize them anew.

m [MeV]	Variance	Bias	Sum	Exact mean (95% CI)	Observed
50	0.989	0.196	1.185	1.179 [1.125, 1.233]	1.096
71	0.994	0.112	1.106	1.095 [1.045, 1.145]	1.925
78	0.996	0.030	1.026	1.022 [0.976, 1.067]	2.330
120	0.998	0.001	0.998	1.008 [0.968, 1.047]	0.780
250	0.998	0.000	0.998	0.992 [0.967, 1.018]	0.820

Table 2: All entries use Q/N_{bin} . The first three columns decompose the fixed- V linear prediction. Exact replay updates V on each toy. Mean intervals are pointwise.

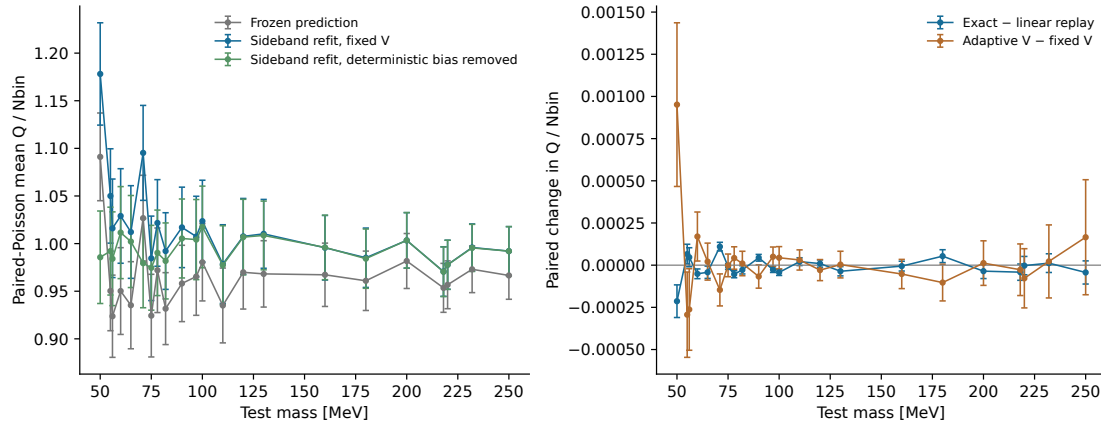


Figure 6: Paired controls isolate sideband-estimator variance and deterministic interpolation response. Subtracting δ is a diagnostic counterfactual; observed counts and the physical source are unchanged. The differences on the right are small compared with the observed curve’s structure.

All 201 masses were replayed on the same 256 spectra (51,456 exact GP predictions). The largest absolute paired mean change from the linear prediction is 2.14×10^{-4} ; updating rather than freezing V changes the mean by at most 9.52×10^{-4} in Q/N_{bin} .

The original slide-14 curve is reproduced exactly. Its maximum is $Q/N_{\text{bin}} = 2.329997$ at 78 MeV, with 16 window bins. The exact conditional mean there is 1.0217 (95% mean interval [0.9762, 1.0672]). Removing the deterministic residual in the fixed- V replay gives mean 0.9903. The total deterministic bias power is 0.487 before division by 16, while the signed-root response squared is $a^2 = 0.333$.

As an exploratory check, define $T_Q = \max_m Q(m)/N_{\text{bin}}(m)$ over the same 201 masses. Twenty of 256 exact paired scans exceed the observed T_Q . The add-one tail is 0.0817 with a 95% binomial interval [0.0484, 0.1181]. This is a conditional residual-scan diagnostic, not a resonance significance or a calibrated physical-background goodness-of-fit probability. It omits source-estimation uncertainty.

6 Consequences for interpretation and the next validation

The study identifies a predictable maximum-selection effect and a genuine conditional fixed-mass response offset. The source-response field and the complete-refit means agree at the tested 2021 precision. The residual replay finds no large covariance-normalization shift capable of explaining the observed feature by itself. These checks do not establish that the frozen source is an adequate physical background everywhere.

The small Q/N_{bin} bias term at 78 MeV and the nonzero signal-root offset are compatible. Q aggregates residual power over many bin directions and divides by the bin count; r measures a coherent, signal-shaped direction after likelihood profiling. A modest coherent residual can affect the signal projection without making the average per-bin residual power large. The two statistics should not be identified.

What the existing slides should communicate

Slide 50 should explicitly distinguish the nominal asymptotic local tail from the source-conditional marginal tail and the full-scan tail. Its maximum histogram should not be centered at zero. Slide 14 should call Q/N_{bin} a descriptive residual statistic, with the unit line labeled as a reference, and ideally show the conditional expected curve. No presentation edits are made as part of this study.

Checks beyond the primary 2021 question

Scope	min a	max a	RMS a	Range of s
Full 2015	-0.735	0.738	0.304	0.929–0.972
Full 2016	-1.394	1.352	0.605	0.927–0.987
2021 native 10%	-1.096	1.400	0.251	0.945–0.987
Shared-coupling union	-1.237	0.819	0.276	0.941–0.987

Table 3: Saved source-response summaries. Mass ranges are 19–100, 39–180, 50–250 and 19–250 MeV, respectively. RMS and ranges are descriptive across correlated mass nodes.

The source offsets persist in other scopes and change sign. A secondary 2016 width check flags adjacent 43.5 and 44 MeV points: direct standard deviations are 1.130 and 1.118 versus response scales 0.955 and 0.954. The model scales lie below the lower bounds of the per-scope Bonferroni normal-theory intervals. These correlated neighbors indicate one unresolved region; they are not two independent failures. This prevents a blanket claim of field-width closure across all datasets. It does not by itself identify a code defect or contradict the reported 2021 comparison.

Priority follow-up

First test the response offset against independently specified smooth background sources, including curvature and boundary variations motivated by the source construction. Repeat the complete masked procedure and inspect signed means, widths and signal injection recovery. This tests whether $a(m)$ is specific to interpolation of the observed-data-derived source or persists over plausible backgrounds. The 50 MeV boundary and 71–78 MeV structure deserve explicit controls. Separately resolve the 2016 width region with additional direct refits before relying on rare tails there.

Keep the production $\pm 2.25\sigma_m$ mask fixed during these controls. Choosing a smoother source, subtracting an observed feature, changing the mask, or recentering a root because it improves an observed score would change the procedure under test. Neither the current conditional agreement nor an attractive corrected plot is a substitute for source uncertainty, injection recovery and coverage studies.

7 Reproduction and provenance

The accompanying archive contains copied numerical inputs, scripts, CSV/JSON results, vector figures, this source, independent checks and a SHA-256 manifest. Parent releases are used as immutable inputs. The primary sources are Analysis Note v5.0.5, the v5.8.2 complete Poisson ensemble and v5.8.5.3 raw-significance fields. The broader v5.9 signal-tail prescription is not varied in this null-only study.

Input	Purpose and identity (SHA-256 prefix)
v5.0.5 spectrum	Observed counts, mass grid, resolution and archived kernels; fd4867a7e2df69d8
v5.8.2 null ensemble	Frozen source and 256 paired complete Poisson spectra; 306a915b9ba6230a
v5.8.5.3 2021 field	Raw roots, source response, scale, correlation, direct validation and saved maxima; 3e444bc3387a3074

All arrays of the v5.0.5 and v5.8.2 saved spectra agree exactly, as do the generating source used by the residual replay and slide-50 calibration. Full hashes and source paths are in the input manifests. Source agreement does not make the observed-data-derived source statistically independent of the observation.

Run order. From the extracted study directory, run `bash reproduce.sh`. Python requires NumPy, SciPy and Matplotlib; the report uses Tectonic. Each numerical script sets one BLAS thread. Read `README.md` for output files, conditional assumptions and the precise scope of the checks. No remote data access is required for the numerical reproduction.

Finite samples. Binomial intervals are two-sided exact Clopper–Pearson intervals. Listed Gaussian-field tail estimates use $(k + 1)/(N + 1)$; raw direct counts are reported separately. Mean intervals use Student’s t and scale intervals use normal-theory chi-square bounds. The latter are diagnostics that assume sufficiently normal marginal roots. Paired differences retain common draws; neighboring mass nodes are not independent experiments. The 200,000-field Monte Carlo precision measures integration error within a frozen model, not uncertainty in that model.

Validation. The package checks source identity, correlation normalization, unchanged raw thresholds, finite-difference derivatives, exact residual moment identities and replay agreement. A separate reviewer audits the scientific definitions and interpretation. The PDF is checked both through extracted text and rendered pages. Detailed results are in `review/` and `qa/`.

References

- [1] G. Cowan, K. Cranmer, E. Gross and O. Vitells, Asymptotic formulae for likelihood-based tests of new physics, Eur. Phys. J. C 71 (2011) 1554, arXiv:1007.1727.
- [2] V. Ananiev and A. L. Read, Gaussian Process-based calculation of look-elsewhere trials factor, JINST 18 (2023) P05041, arXiv:2206.12328.
- [3] C. E. Rasmussen and C. K. I. Williams, Gaussian Processes for Machine Learning (MIT Press, 2006), chapter 2, Gaussian-process regression.